CHEM1043 General Chemistry II
Assignment 2 (Topics 3 – 4)

Assignment 2 (15% of total course mark) is released on 14 Oct 2026 (Wed). Please put your work on A4- sized white paper (blank or lined) and scan your ***handwritten*** *work into one single pdf file. Name the file in the format of 10432B_<your student number>, e.g. 10432B_3035000000. Submit the file through* ***Moodle Assignment by 8:00 pm on 11 Nov 2026 (Wed).***

Please refer to your lecture notes and textbook, and the list of physical constants and the periodic table posted on Moodle for data/constants/equations as necessary.

Section A: Solutions and their properties [47 marks]

1. The aqueous solubility of CO_2 at 20°C and 1.00 atm is equivalent to 87.8ml CO_2(g), measured at 0 °C and 1.00 atm, per 100ml of water. What is the molarity of CO_2 in water that is at 20° C and saturated with air at 1.0 atm? The volume percent of CO_2 in air is 0.0360%. Assume that the volume of the water does not change when it becomes saturated with air.

(6 marks)

2. What are colligative properties? (6 marks)

3. Calculate the boiling and freezing points of an aqueous solution containing 60 g glucose, $C_6H_{12}O_6$, dissolved in 800 mL water? Take the density of water as 1.00 g/mL; its $K_b = 0.52$ ° C·kg·mol^{-1} and its $K_f = 1.86$ °C·kg·mol^{-1}.

(6 marks)

4. A solution is made by dissolving 0.0653 kg of biphenyl ($C_{12}H_{10}$) in 300.0 mL of benzene (C_6H_6).

a) If the vapor pressure of pure benzene is 35.7 kPa at 50.0 °C, then what will the vapor pressure of the solution be? (Consider biphenyl to be nonvolatile and the density of benzene is 0.877 g/mL)

(4 marks)

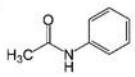
b) What would be the osmotic pressure of this solution at 40.0 °C? (The density of benzene is 0.877 g/mL.). (4 marks)

5. Most gasoline sold in the United States contains ethanol (C_2H_5OH) in addition to mixture of hydrocarbons such as octane (C_8H_{18}). At 20°C, the vapor pressure of ethanol is 45 torr and the vapor pressure of octane is 10 torr. What is the vapor pressure at 20°C of a solution prepared by mixing 8.16g ethanol and 63.3g octane?

(6 marks)

6. Consider using recrystallization to purify a 4.5 g sample of A, which has been contaminated with 0.2 g of B, by decreasing temperature from 100 to 20 °C.

(9 marks)

	A Salicylic acid 	B Acetanilide 
Solubility in water at 100 °C	7.5 g / 100 mL	
Solubility in water at 20 °C	0.20 g / 100 mL	0.5 g / 100 mL

(a) What is the minimum volume of hot water (at 100 °C) required to purify the sample?

(b) How many grams of salicylic acid can be recovered from this recrystallization?

(c) What is the percent recovery?

(d) Suggest two more ways to increase the yield of the recrystallization?

7. A solution contains glucose in 600.00 g of water at 25℃. The vapor pressure of pure water at 25℃ is 23.760 mmHg. If the vapor pressure of the water with glucose is 18.850 mm Hg, how many moles of glucose does this solution contain?

(6 marks)

Section B: Solubility and complex-ion equilibria [42 marks]

8. Consider a solution in which 0.1 mol of each of the following compounds is dissolved in 1 L of water: KCl (potassium chloride), CH_3OH (methanol), NH_4OH (ammonium hydroxide), and $CaCl_2$ (calcium chloride).

Rank the solutions in order of increasing electrical conductivity and please explain your answer.

(6 marks)

9. Classify each of the following into a true solution and a colloidal solution, and explain your answer:
starch solution, salt solution, smoke, ink, sugar solution, vinegar, milk, soap solution.

(6 marks)

10. (a) Calculate $[Ca^{2+}]$ in a saturated solution of CaF_2 ($K_{sp} = 3.9 \times 10^{-11}$).

(b) What mass of NaF must be added to 0.500 L of the solution to decrease $[Ca^{2+}]$ to 1.0×10^{-4}?

(8 marks)

11. An aqueous solution at room temperature is 0.25 M in both Ba^{2+} and Ca^{2+} ions. You want to separate these two ions by selective precipitation using carbonate. What percentage of the first ion will have precipitated when the second ion begins precipitating? ($BaCO_3$ Ksp = 2.58×10^{-9} M, $CaCO_3$ Ksp = 3.36×10^{-9} M).

(6 marks)

12. Why does the addition of HCl or NaOH to a saturated solution of $Al(OH)_3$ in contact with solid $Al(OH)_3$ increase the solubility of the solid?

(5 marks)

13.Sometimes equilibria for complex ions are described in terms of dissociation constants, K_d.

For the complex ion $Zn(CN)_4^{2-}$, the dissociation reaction is as follows.

$Zn(CN)_4^{2-} \longrightarrow Zn^{2+} + 4CN^-$ $K_d = [Zn^{2+}] \cdot [CN^-]^4 / [Zn(CN)_4^{2-}] = 2.4 \times 10^{-20}$

Calculate the value of the formation constant, K_f for $Zn(CN)_4^{2-}$.

(5 marks)

14. Will the following precipitates form under the given conditions? (6 marks)

(a) $PbI_2(s)$, from a solution that is 1.05×10^{-3} M HI, 1.05×10^{-3} M NaI, and 1.1×10^{-3} M $Pb(NO_3)_2$

(b) $Mg(OH)_2$ (s), from 2.50 L of 0.0150 M $Mg(NO_3)_2$ to which is added 1 drop (0.05ml) of 6.00 M NH_3.

(PdI_2 $K_{sp} = 7.9 \times 10^{-9}$, NH_3 $K_b = 1.8 \times 10^{-5}$, $Mg(OH)_2$ $K_{sp} = 5.6 \times 10^{-12}$)